



Antenna Datasheet

Product OC: YFGA018E3AM

Version: 2.3

Date: 2025-10-27

Status: Released

Product Name: GNSS Buckle Mount Ceramic Patch + Cable Active
Embedded Antenna

Key Features:

Frequency Band: 1559–1606 MHz

Dimensions: 18 mm × 18 mm × 7.5 mm

LNA Gain: 16 ±3 dB

RoHS and REACH Compliant

Overview

The Quectel YFGA018E3AM is a high-performance GNSS buckle mount ceramic patch + cable active embedded antenna, designed for reliable positioning. Operating in the 1559–1606 MHz range with 50 Ω impedance and RHCP polarization, it supports key bands like GPS L1, GLONASS G1, GALILEO E1, BDS B1I/B1C, and QZSS L1.

With dimensions 18 mm × 18 mm × 7.5 mm and a 7.7 g weight, it features a Φ 1.13 mm black RF cable (100 mm) with an IPEX MHF1 connector and buckle mounting. Its LNA gain of 16 $\pm$ 3 dB and noise figure $\leq$ 2.5 dB enhance signal reception, complemented by 51 % efficiency at 1561 MHz and a peak gain of -0.46 dBi.

RoHS and REACH compliant, it operates in -40 °C to +85 °C. Packaged in pearl cotton trays (36 pcs) and cartons (180 pcs), it suits seamless integration in GNSS-enabled devices requiring robust performance in diverse environments.

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Free Space

1.1. Electrical

Electrical	
Frequency Range	1559–1606 MHz
Impedance	50 Ω
Polarization	RHCP
Radiation Pattern	Directional

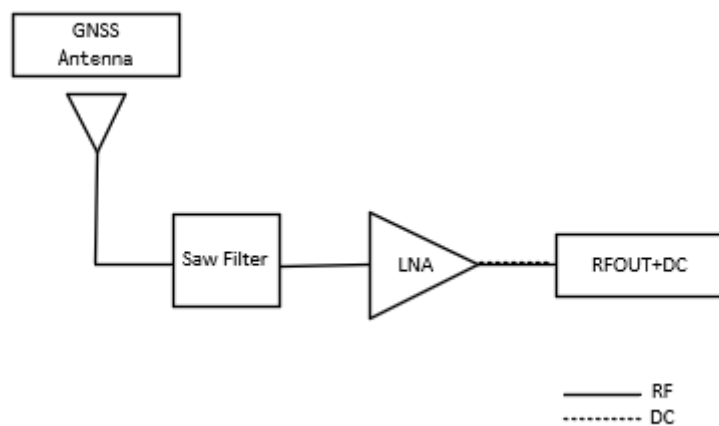
Frequency (MHz)	Band	GPS L5 GALILEO E5a BDS B2a- B2I QZSS L5 IRNSS L5	GALILEO E5b BDS B2b	GPS L2 QZSS L2C	GLONASS G2	BDS B3	BDS B1I	GPS L1 GALILEO E1 BDS B1C QZSS L1	GLONASS G1
		1176	1207	1227	1248	1268	1561	1575	1602
VSWR		-	-	-	-	-	1.46	1.6	12.3
Return Loss (dB)		-	-	-	-	-	-14.5	-12.7	-1.4
Efficiency (%)		-	-	-	-	-	51	44	8.4
Peak Gain (dBi)		-	-	-	-	-	-0.46	-0.3	-7

LNA Electrical	
LNA Gain	16 ±3 dB
Noise Figure	≤ 2.5 dB
Output VSWR	< 2.0
Filter Out-of-Band Attenuation	≥ 45 dB f0 ±100 MHz f0 (1568 MHz)
Working Voltage	1.8–3.3 V
Working Current	4.3 ±1.5 mA
Impedance	50 Ω

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	18 mm × 18 mm × 7.5 mm
Material	PCBA + Ceramic + RF Cable
Cable Type & Color & Length	Φ 1.13 mm & Black & 100 mm
Connector Type	IPEX MHF1
Mounting Type	Buckle
Weight	Typ. 7.7 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS & REACH Compliant	Yes

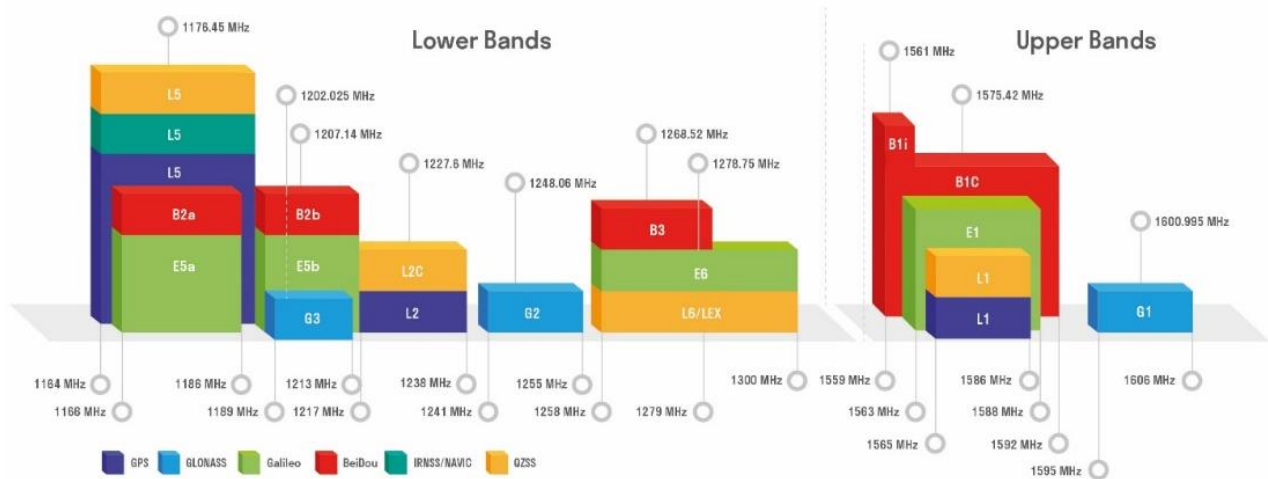
1.3. Block Diagram (Active Antenna)



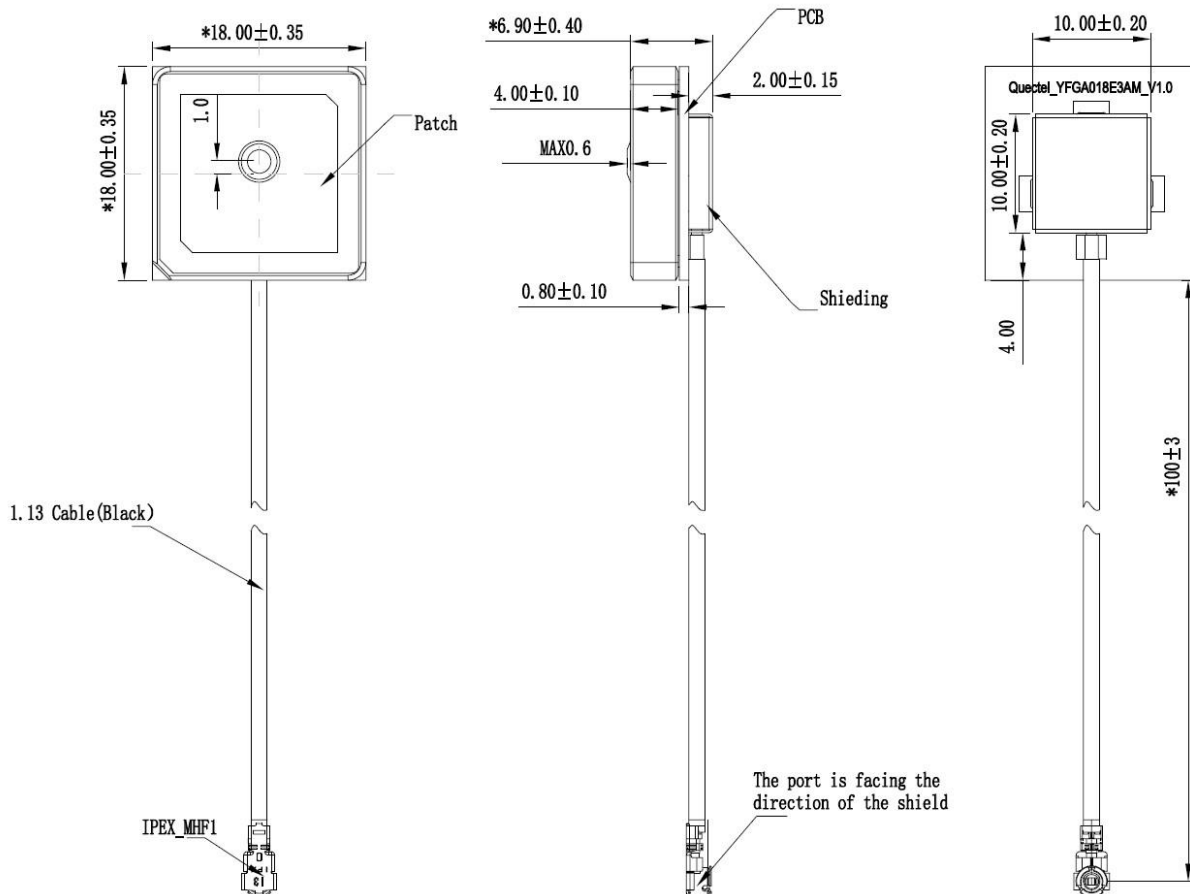
1.4. Supported GNSS Frequency Bands

GNSS Frequency Bands (MHz)					
GPS	L1 Centre 1575.42 (1565–1586)	L2 Centre 1227.6 (1217–1238)	L5 Centre 1176.45 (1164–1189)		
	√	-	-		
GLONASS	G1-L10C-L10F Centre 1601 (1595–1606)	G2-L20C-L20F Centre 1248.06 (1241–1255)	G3-L30C Centre 1202.025 (1189–1213)		
	√	-	-		
GALILEO	E1 Centre 1575.42 (1563–1588)	E5a Centre 1176.45 (1166–1187)	E5b Centre 1207.14 (1197–1218)	E6 Centre 1278.75 (1258–1300)	
	√	-	-	-	
BDS	B1I Centre 1561.098 (1559–1564)	B1C (BDS-3) Centre 1575.42 (1559–1592)	B2a-B2I Centre 1176.45 (1166–1187)	B2b Centre 1207.14 (1197–1217)	B3 Centre 1268.52 (1258–1279)
	√	√	-	-	-
QZSS	L1 Centre 1575.42 (1573–1578)	L2C Centre 1227.6 (1226–1229)	L5 Centre 1176.45 (1166–1187)	L6 Centre 1278.75 (1257–1300)	
	√	-	-	-	
IRNSS	L5 Centre 1176.45 (1164–1189)				
	-				

GNSS Bands and Constellations



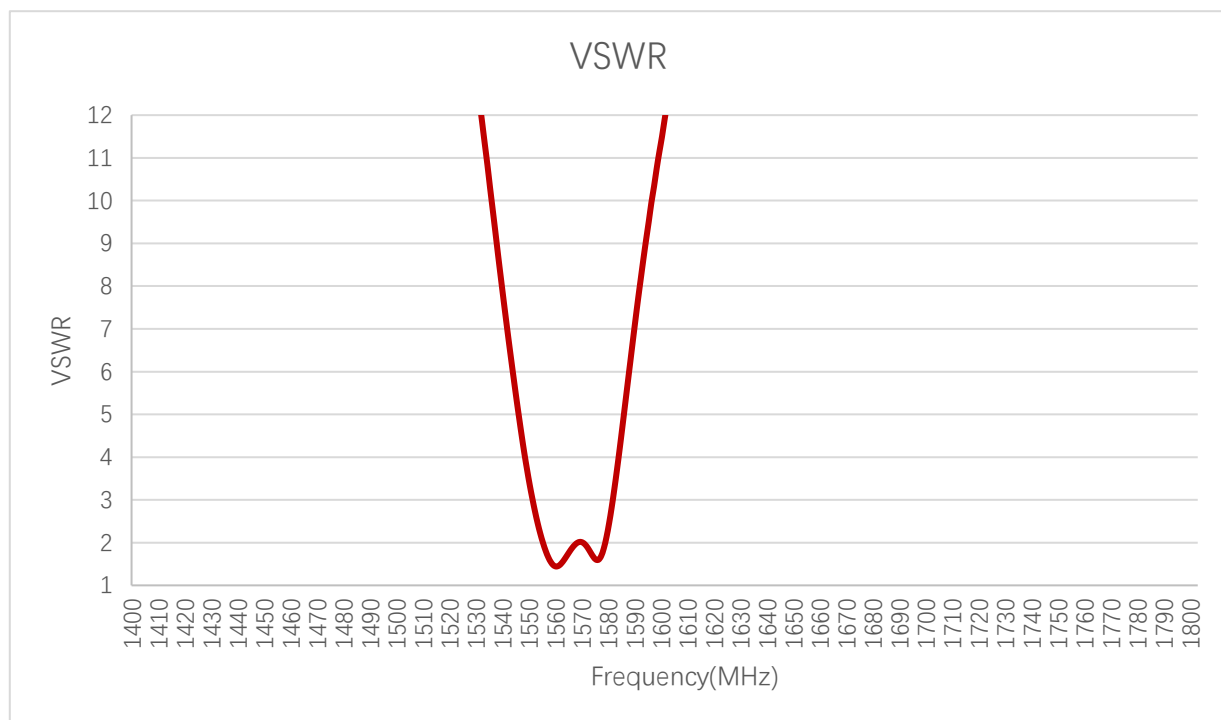
2 Drawing



3 Detailed Performance

3.1. S-Parameter Test

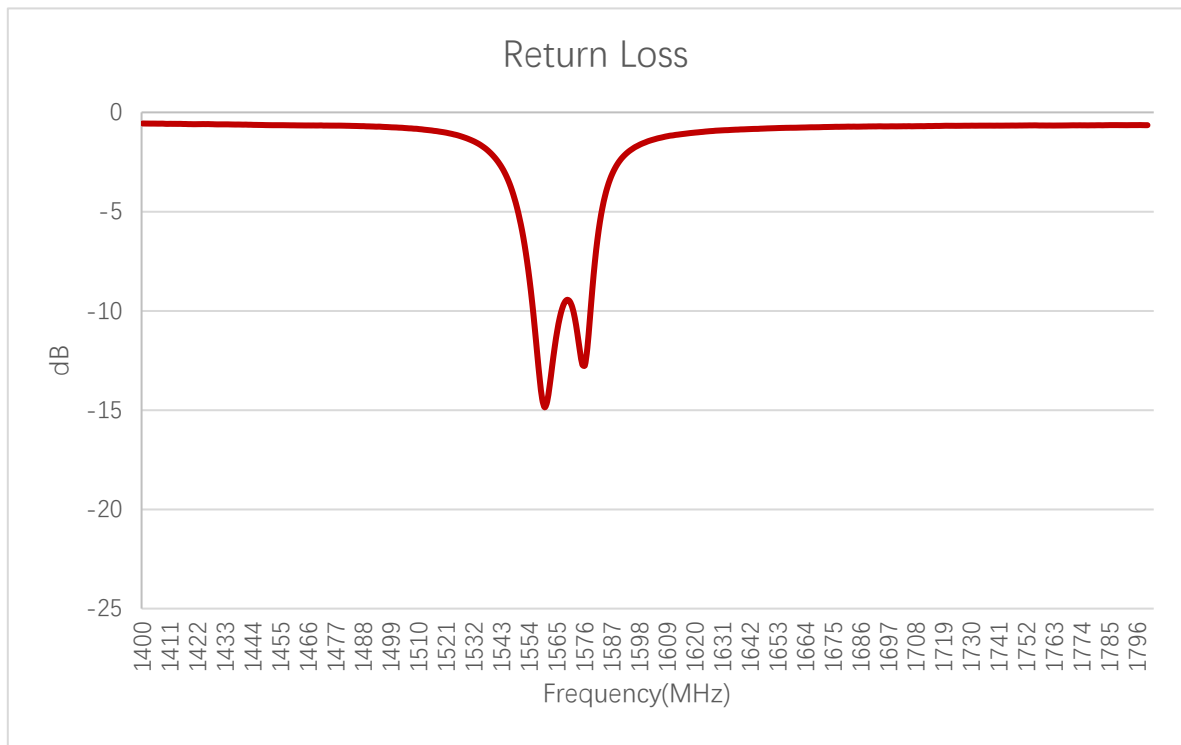
3.1.1. VSWR



VSWR

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
VSWR	-	-	-	-	-	1.46	1.6	12.3

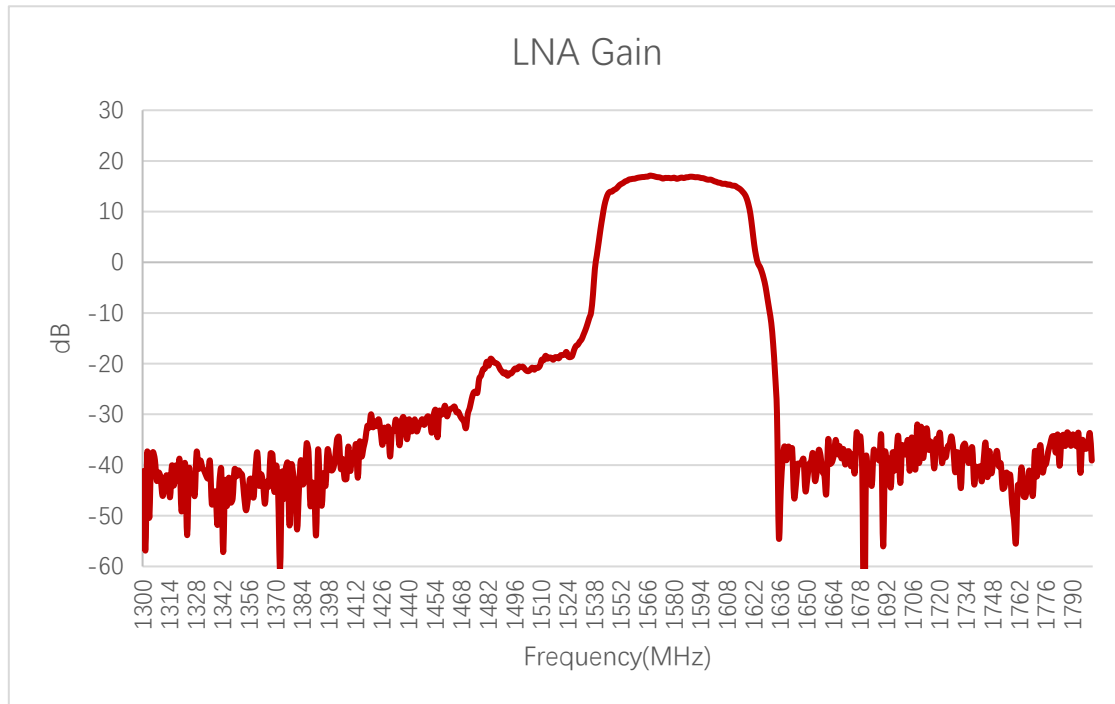
3.1.2. Return Loss



Return Loss (dB)

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Return Loss (dB)	-	-	-	-	-	-14.5	-12.7	-1.4

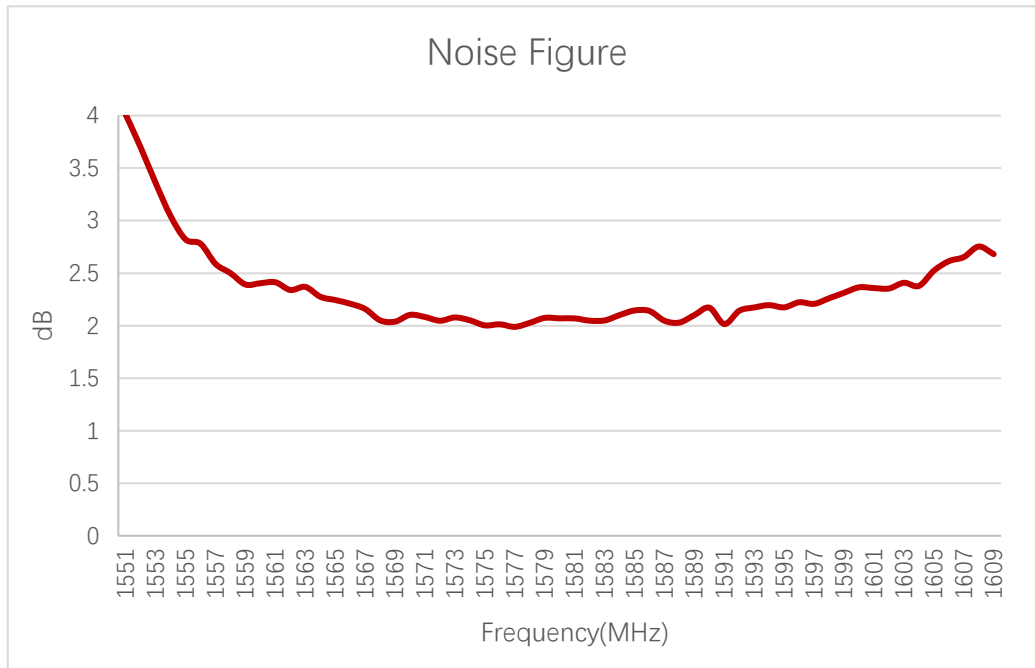
3.1.3. LNA Gain



LNA Gain (dB)

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
LNA Gain (dB)	-	-	-	-	-	16.7	16.6	15.8

3.1.4. Noise Figure

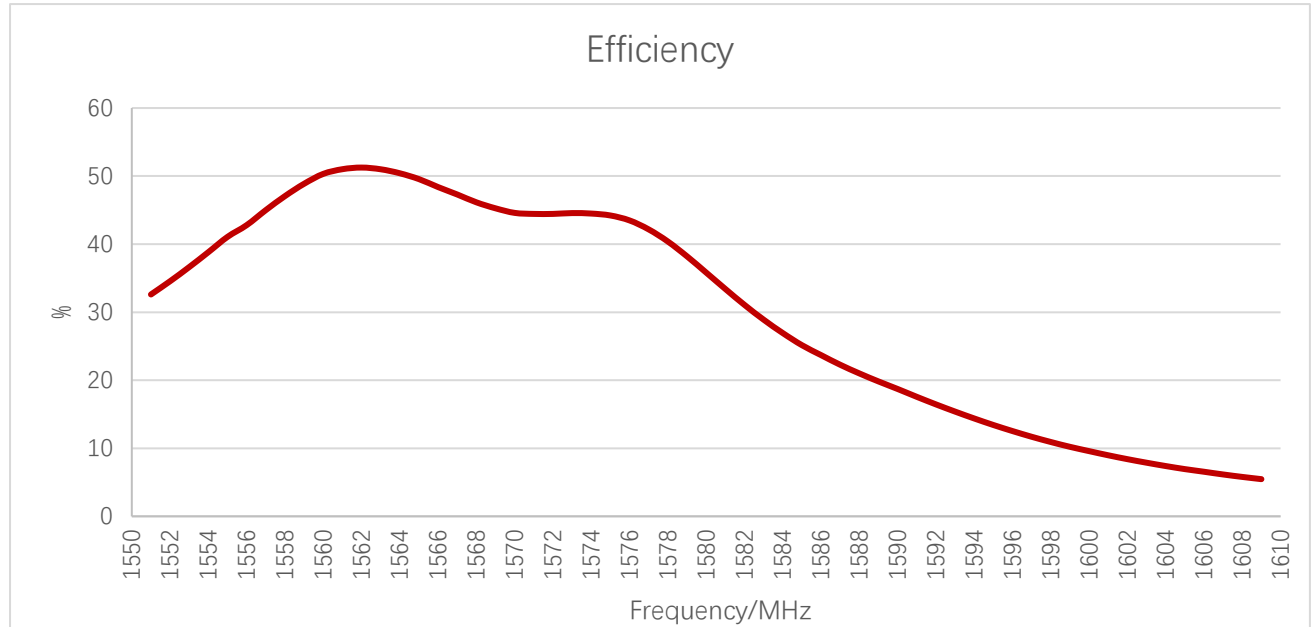


Noise Figure (dB)

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Noise Figure (dB)	-	-	-	-	-	2.4	2	2.35

3.2. Radiation Performance Test

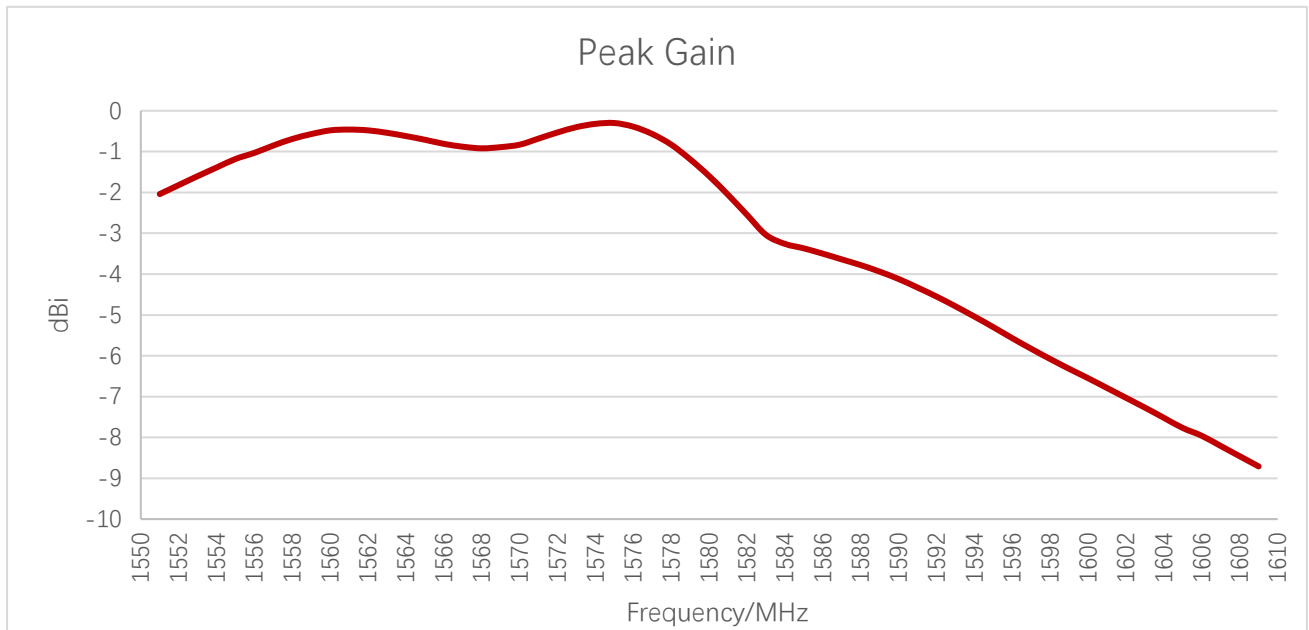
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Efficiency (%)	-	-	-	-	-	51	44	8.4

3.2.2. Peak Gain

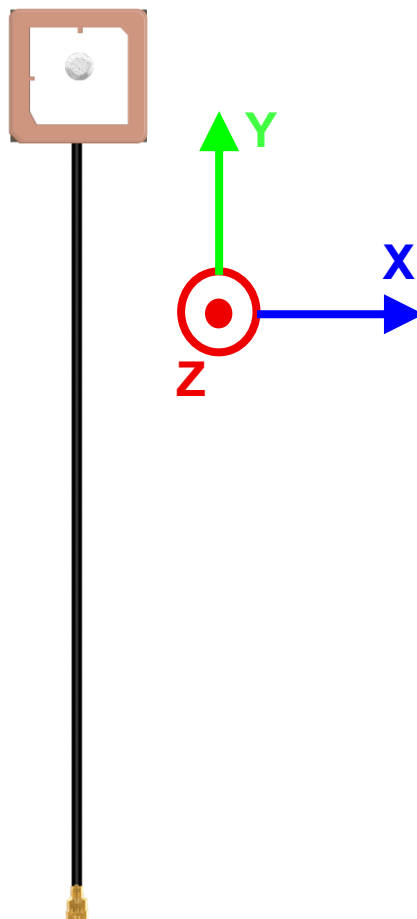


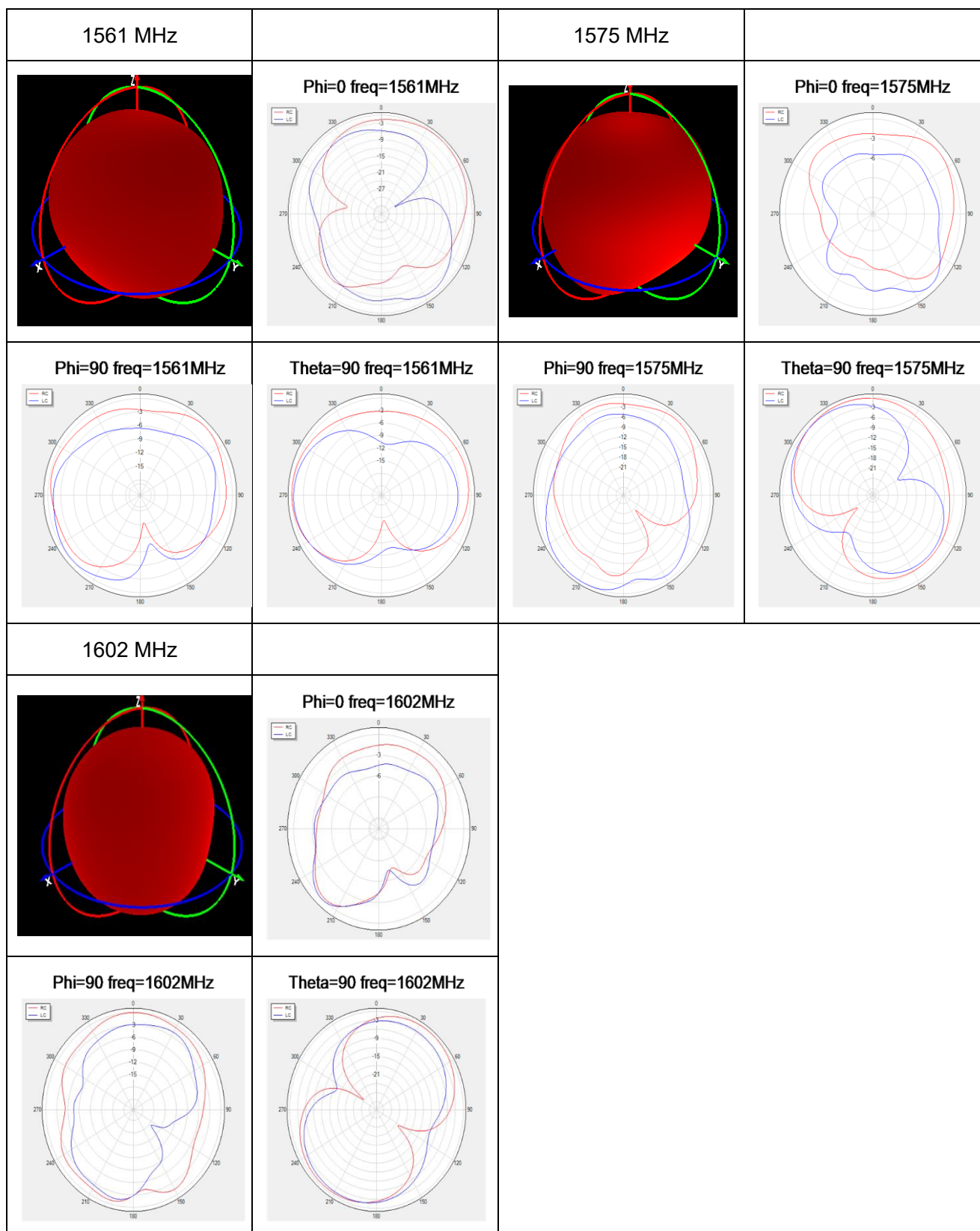
Peak Gain (dBi)

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Peak Gain (dBi)	-	-	-	-	-	-0.46	-0.3	-7.0

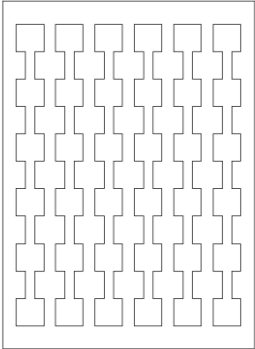
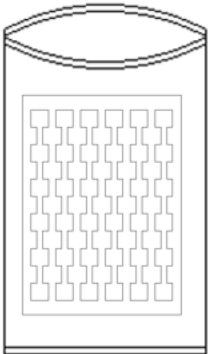
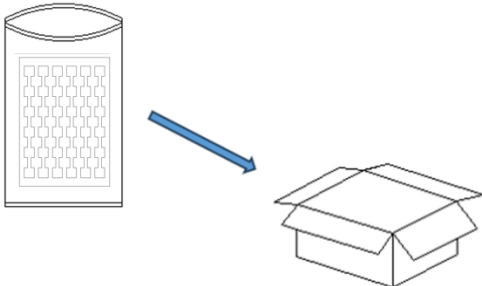
3.2.3. 3D & 2D Radiation Pattern

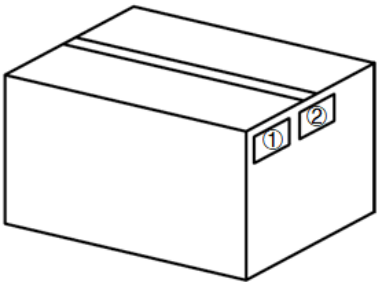
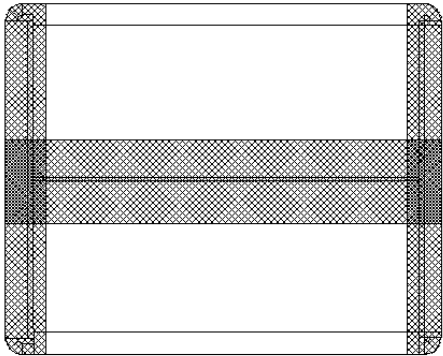
- Test Condition: Free Space
- Test Chamber: SH-SY-16M





4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		(36 Antennas / Pearl Cotton Tray)
2		The pearl cotton tray is vacuumed in a vacuum bag.
3		(5 Pearl Cotton Trays / Carton Box) (180 Antennas / Carton Box) <u>Carton Size:</u> <u>L × W × H = 405 × 293 × 185 mm</u>

4		Position for Attaching Labels ① Carton Label ② Quality Label
5		Sealing Cartons H-shaped sealing cartons

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Or our local offices. For more information, please visit:

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Revision History

Version	Date	Author	Note
-	2024-04-18	Junsen Li/ Poker Guo/ David Liu/ Rainey Liao	Creation of the document
1.0	2024-04-18	Junsen Li/ Poker Guo/ David Liu/ Rainey Liao	First official release
2.0	2024-12-07	Rhone Wei	Numerous changes were made to this document. It should be read in its entirety.
2.1	2024-12-26	Mike Guo	Deleted adhesive mounting type of antenna (Chapter 1.2).
2.2	2025-07-14	Aria Chu	1. Updated the antenna image (Cover page). 2. Updated the overview.
2.3	2025-10-27	Junsen Li	1. Added the LNA gains according to different supply voltages (Chapter 1.1). 2. Updated the figure (Chapter 1.3).



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